✈ WayFinder

The AI travel strategist that reads the traveler, not just the query

Expedia Hackathon · Problem Statement 1 · AI Air Travel Companion

Solo submission — Tarun Aryan

50,000 itineraries · 50 travelers · 6/6 judge benchmarks · 42/42 expected behaviors verified

Same question, six different right answers

> "I want to visit Sydney around the holidays."

- For **U05** (Lisbon, First cabin, 90-min layover cap): a Business cabin via Paris at **$10k** — with an honest warning that no First is sold on the route.
- For a broke student, the right answer is **two stops and $900**.

Flight search that ignores *who's asking* is broken. WayFinder fuses each traveler's **structured profile** with their **messy free-text booking history** into constraints, weights, and receipts.

Given: flights_data.csv (50,000 rows) · user_data.csv (50 travelers) · benchmark_prompts.json (6 judged prompts)

The data has traps — we found them and built for them

Trap in the dataset	What WayFinder does
LIS→SYD has zero direct flights, no First cabin — while U05 demands direct-only + First	Route-reality banners + a relaxation ladder that narrates every constraint it loosens
Service is sparse by month (MEL↔JFK never pairs Tue–Thu)	Self-composed connections & nearest-window scan across 6 months
Deliberate contradictions: age 66 vs *"broke student"* in history	Contradiction detection — shown to the user, resolved by behavioral-evidence-wins
Histories are messy free text (*"redeyes kill my mornings"*)	Preference mining with provenance: every chip cites its source

Architecture — deterministic core, AI-assisted edge

```
query ─► NLU (travel clock · gazetteer · trip shape)
      ─► Preference Fusion (structured ⊕ mined history → provenance-tagged)
      ─► Search (time-expanded graph · beam search · permutation chains ·
                 self-composed connections · relaxation ladder)
      ─► Ranking (profile-derived weights → 0–100 fit score + breakdown)
      ─► Insights (Recommended/Cheapest/Fastest deltas · seasonal premiums)
      ─► Narrative (evidence-cited) ─► conversational refinement loop
```

- **AI where it helps:** local LLM (Ollama) parses unusual phrasing & polishes prose — behind a **number-integrity check** that rejects any hallucinated figure.
- **Guarantee:** with the LLM completely off, all 42/42 benchmark behaviors still pass. *The demo cannot die on stage.*

Preference fusion with receipts

WayFinder
reads the traveler, not just the query

2025-08-01 | 50,000 flights · 1,172 routes | AI: deterministic engine

Filter 50 travelers...

- 01 **Cape Town** U01 B01 — Business · Business · Low Price Sens.
- 02 **Mexico City** U02 B02 — Leisure · Economy · High Price Sens.
- 03 **Amsterdam** U03 B03 — Leisure · Economy · Medium Price S...
- 04 **Melbourne** U04 B04 — Mixed · Economy · High Price Sens.
- 05 **Lisbon** U05 B05 — Leisure · First · None Price Sens.
- 06 **Chennai** U06 B06 — Leisure · Economy · High Price Sens.
- 07 **Delhi** U07 — Leisure · Premium Economy · Mediu...
- 08 **Rio de Janeiro** U08 — Business · Business · Medium Price ...
- 09 **Sydney** U09 — Mixed · Economy · Medium Price Se...
- 10 **Rome** U10 — Leisure · Economy · Medium Price S...
- 11 **Moscow** U11 — Business · Business · Low Price Sens.
- 12 **Amsterdam** U12 — Leisure · Economy · High Price Sens.
- 13 **Moscow** U13 — Leisure · Economy · Medium Price S...
- 14 **Seoul** U14 — Mixed · Economy · High Price Sens.
- 15 **Rio de Janeiro** U15 — Leisure · First · None Price Sens.

Plan a trip | Benchmark self-grading

Lisbon (LIS) U05 · age 24 — mined from history · profile column

HARD CONSTRAINTS
- max layover minutes: **90**

STRONG PREFERENCES
- direct preference: **strong**
- price sensitivity: **none**
- preferred cabin: **First**
- seasonal months: **[6,7,8,12]**
- budget priority: **none**

SOFT SIGNALS
- preferred airlines: **["LH"]**
- baggage: **{"carryon_only":false,"che...**
- frequent flyer: **Miles & More**
- peak season ok: **true**
- premium services wish: **spa lounge, chauffeur tran...**

▸ Raw booking history (4 notes — the messy data we mine)

I want to visit Sydney around the holidays — what should I expect? | Plan it

Judge benchmarks — one click selects the right traveler and runs their exact prompt:

- **B01** I need to get from home to Tokyo next month, what do you sug...
- **B02** Find me the best way to do a London + Paris + Rome trip in o...
- **B03** Cheapest option to Bali, I'm flexible on dates over the summ...
- **B04** Book me something to New York for a Tuesday meeting, back Th...
- **B05** I want to visit Sydney around the holidays — what should I e...
- **B06** Plan a multi-city Asia trip, I have about three weeks of fle...

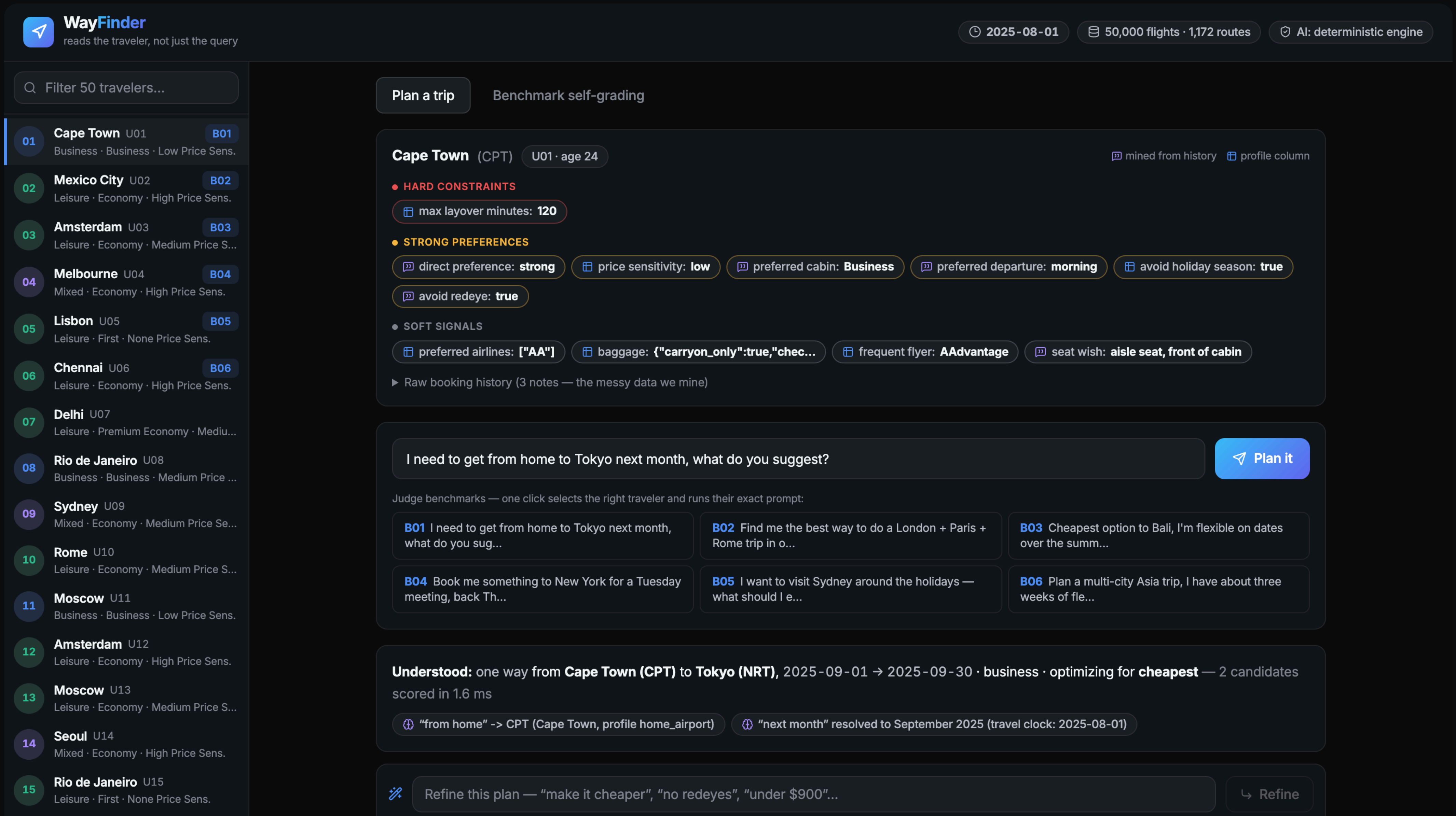
Understood: one way from **Lisbon (LIS)** to **Sydney (SYD)**, 2025-12-11 → 2026-01-09 · **leisure** — 1 candidates scored in 6.8 ms

- origin defaulted to home airport LIS (Lisbon)
- "around the holidays" -> year-end window 2025-12-15..2026-01-05

Refine this plan — "make it cheaper", "no redeyes", "under $900"... | Refine

make it cheaper | no redeyes | direct only | under $1,000 | a week later

Optimization that explains itself — and takes follow-ups



Graded by the judges' own rubric — live, in the app

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14	**Seoul** U14		Mixed · Economy · High Price Sens.
15	**Rio de Janeiro** U15		Leisure · First · None Price Sens.
16	**Rio de Janeiro** U16		Leisure · Economy · High Price Sens.

Plan a trip | **Benchmark self-grading**

Runs the judges' own **benchmark_prompts.json** live and verifies every *expected_behavior* programmatically against the actual response — the system grades itself on the official rubric.

Run all 6

42/42 behaviors verified · **37** ms avg response · **6/6** prompts run

B01 U01 "I need to get from home to Tokyo next month, what do you suggest?" — 7/7

Infer preferences from BOTH structured fields and raw_history	profile highlights draw on ['raw_history', 'structured']; e.g. raw: "always book business, hate connections"
Respect direct_preference='strong' and max_layover_minutes=120	cap=120min; all 2 recommendations within cap
Weight cost vs convenience per price_sensitivity='low'	derived weights: {'price': 0.1579, 'convenience': 0.3684, 'time': 0.2105, 'reliability': 0.1579, 'preffit': 0.1053}; price weight 0.15 ← price_sensitivity='low'
Depart from home_airport=CPT; airline preferences engaged	all itineraries start at CPT; preference-fit feature scored (preferred: ['AA'])
Surface the cost-vs-time trade-off explicitly	Our recommended pick is also the cheapest available.
Account for seasonal/holiday pricing and seat scarcity	Heads-up: no Business cabin is offered on this route for these dates (available: Economy) — showing the closest cabins instead.
Explain WHY the top pick fits this traveler, citing the evidence used	2 evidence-cited reasons; raw-history quoted: True; e.g. "Non-stop routing matches your direct-flight preference" ← "always book business, hate connections"

▸ Full narrative · 2 candidates · 2.3 ms

B02 U02 "Find me the best way to do a London + Paris + Rome trip in one journey." — 7/7

Infer preferences from BOTH structured fields and raw_history	profile highlights draw on ['raw_history', 'structured']; e.g. raw: "cheapest fare wins, dont care about stops"
Respect direct_preference='none' and max_layover_minutes=420	cap=420min; zero results under it — relaxation ladder applied and narrated: no feasible city chain in the requested window — scanned the next 6 months for the earliest well-priced chain
Weight cost vs convenience per price_sensitivity='high'	derived weights: {'price': 0.5238, 'convenience': 0.1429, 'time': 0.1905, 'reliability': 0.0952, 'preffit': 0.0476}; price weight 0.55 ← price_sensitivity='high'
Depart from home_airport=MEX; airline preferences	all itineraries start at MEX; preference-fit feature scored (preferred: ['SQ', 'BA'])

Impact & what's next

For travelers — constraints respected (family seats, layover caps, school holidays), expectations set (holiday premiums, seat scarcity), every recommendation explained in their own words.

For the platform — higher search-to-book conversion, fewer support escalations, and an **auditable ranking system**: every decision traces to evidence, not a black box.

Next: live NDC/GDS feeds · learning-to-rank from booking outcomes · hotel/car bundling · CO_2-aware scoring · per-user weight learning from accept/reject feedback.

Demo video

3–5 minutes · the working prototype end-to-end

Same question, different travelers → evidence-cited picks → conversational refinement → relaxation ladder honesty → 42/42 live self-grading

Repository includes README quick start, assumptions, limitations & future work · deterministic reproduction of every claim in this deck